



FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-03 Targeted Remediation Cards

Use only the competency actually failed; then do a changed micro-retest.

Artifact	C3-03-REPAIR
Status	2026 - LEARNER EDITION
Phase	C3-03 Orchestration & Reliable Workflows 13 lessons 34-46 focused hours
Prerequisite	C3-02 Gate PASS; C2B Python/quality/ownership foundations assumed, orchestration taught here.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: simple mental picture -> tiny example -> guided Airflow/tool workflow -> break/fix -> independent changed task -> validation -> production reasoning -> evidence. A green scheduler box alone never proves correct data.



Repair rules

Do not restart the phase for one narrow weakness. Repair the failed competency, close the review, then prove the skill on a changed micro-case. Completing repair never auto-assigns Gate B/C.

R-ARCH - DAG/task boundaries

Redraw a different workflow with input/output/side-effect/retry boundaries; validation must block publish.

PASS: changed evidence is correct, reproducible and explainable with review closed. Do not repeat the original Gate case.

R-TIME - Schedule/data interval

Use a different historical date; derive input/output from data interval with wall-clock date prohibited.

PASS: changed evidence is correct, reproducible and explainable with review closed. Do not repeat the original Gate case.

R-CONFIG - Params/config/connections/secrets

Reclassify fresh values and run a sanitized repo scan; move environment without code changes.

PASS: changed evidence is correct, reproducible and explainable with review closed. Do not repeat the original Gate case.

R-FAIL - Retries/timeouts/failure class

Classify three new failures; run one fail-once transient and one deterministic permanent defect.

PASS: changed evidence is correct, reproducible and explainable with review closed. Do not repeat the original Gate case.

R-REPLAY - Idempotence/backfill

Replay a different old partition after newer entity state exists; prove no regression/duplicates.

PASS: changed evidence is correct, reproducible and explainable with review closed. Do not repeat the original Gate case.

R-WAIT - Sensors/readiness

Implement a different bounded readiness condition and prove ready + timeout paths.

PASS: changed evidence is correct, reproducible and explainable with review closed. Do not repeat the original Gate case.

R-RESOURCE - Pools/concurrency

Protect another scarce dependency with justified pool/active-run limit.

PASS: changed evidence is correct, reproducible and explainable with review closed. Do not repeat the original Gate case.

R-TEST - Testing/deployment

Introduce a fresh import/test defect and prove predeploy checks catch it before scheduling.

PASS: changed evidence is correct, reproducible and explainable with review closed. Do not repeat the original Gate case.

R-OPS - Troubleshooting/runbook

Diagnose a changed incident from run/task/interval evidence; recover minimally and reconcile.

PASS: changed evidence is correct, reproducible and explainable with review closed. Do not repeat the original Gate case.